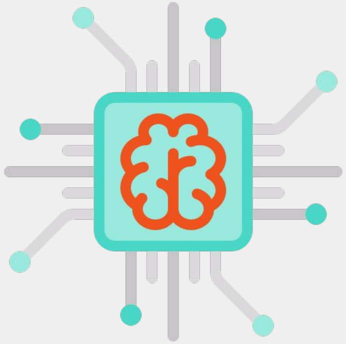
UFSM00291



Lab06

netlist simulation and VCD power analysis

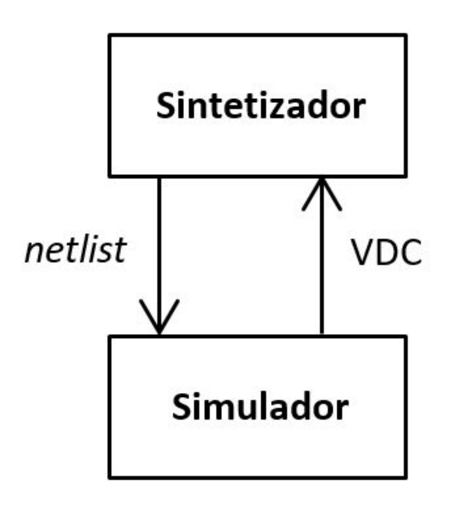
UFSM

Lab 06

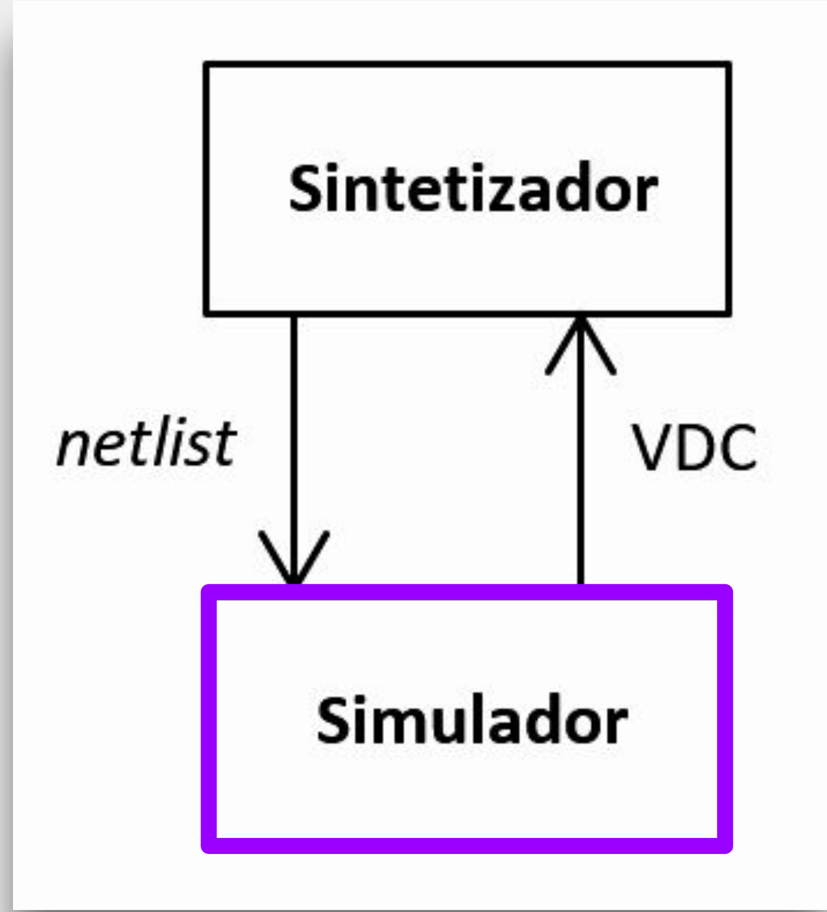
Objetivo: compreender o fluxo para uma simulação com o *netlist* obtido a partir da síntese lógica e a execução da análise de potência.

Pré-requisito(s): labs anteriores
Duração: 2 horas
Suporte:

documentação da(s) ferramenta(s)



Lab 06



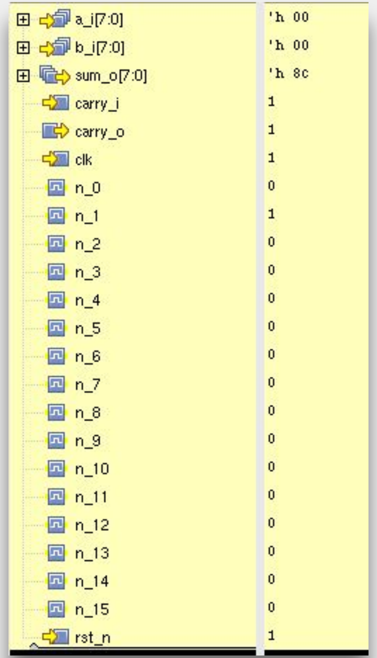
cadence
GENUS
SYNTHESIS SOLUTION
GENUS PHYSICAL SYNTHESIS FLOW

cadence
XCELIUM
Simulation
XCELIUM SIMULATOR

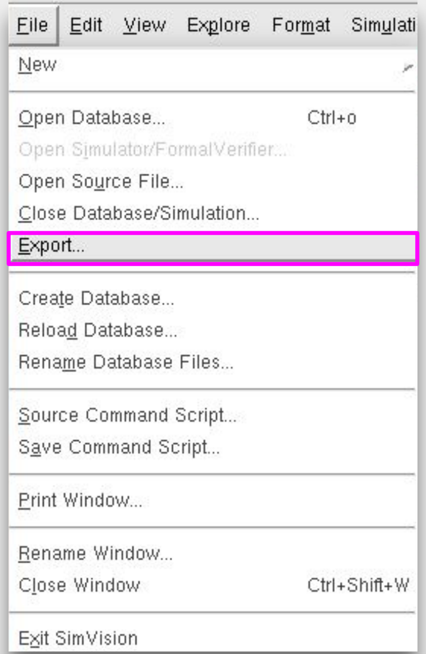
Lab 06

Obtenção do VCD (Xcelium)

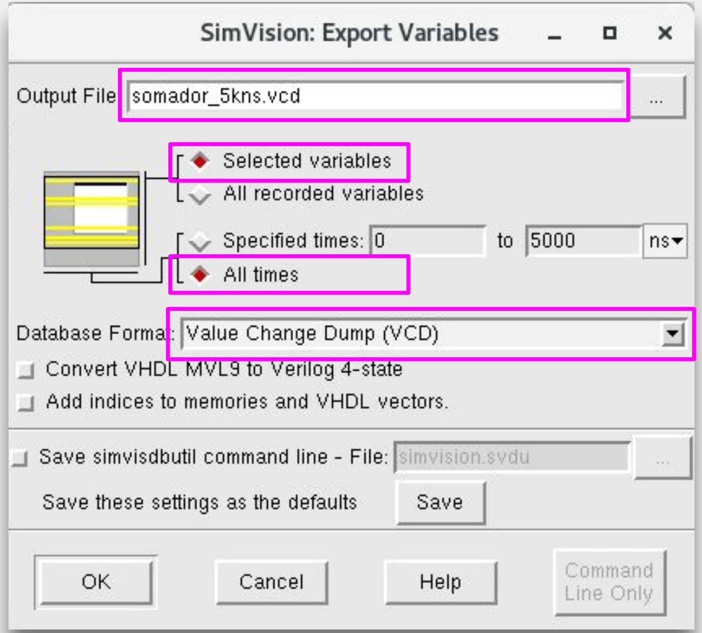
Selecione todas as entradas, saídas e *nets*:



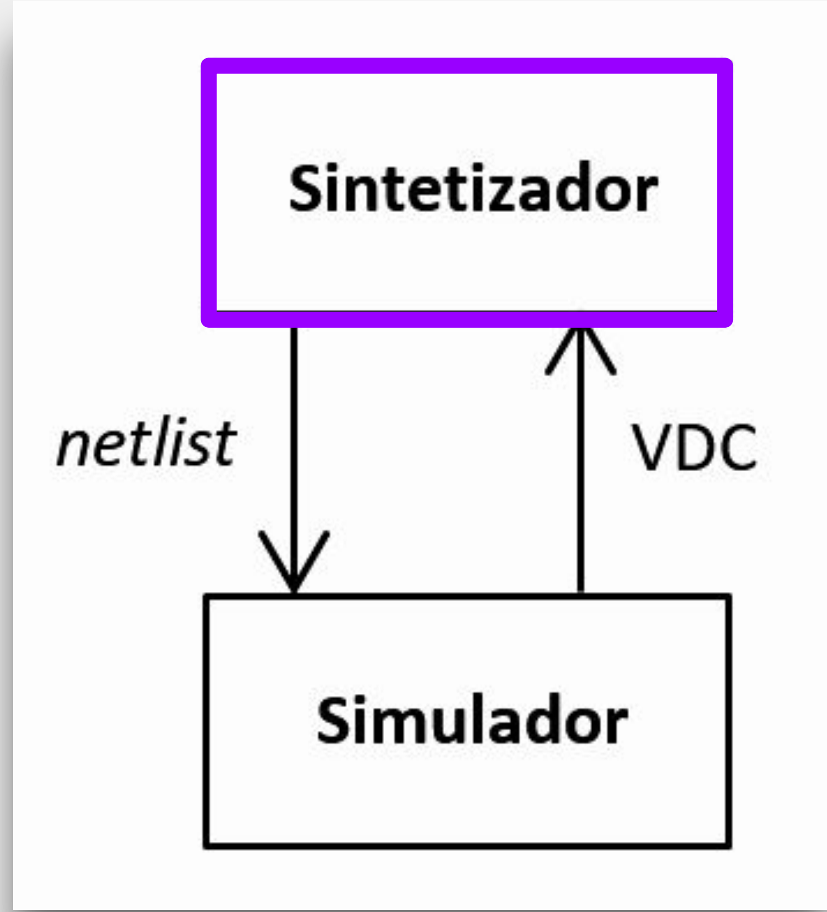
Dentro do Simvision e após executar a simulação, acesse ao menu abaixo para exportar o VCD:



Preencha os dados e clique em OK:



Lab 06



cādence
GENUS
SYNTHESIS SOLUTION
GENUS PHYSICAL SYNTHESIS FLOW

cādence
XCELIUM
Simulation
XCELIUM SIMULATOR

Lab 06

Leitura do VCD e análise de potência (GENUS)

Comando(s) para o arquivo “${DESIGNS}.tcl”:

read_stimulus -allow_n_nets -format vcd -file <arquivo.vcd>

```
---------------------------- Annotation Report ----------------------------
Object Type           Asserted  UnAsserted  Unconnected  Total  Asserted%
                                             + Constant
---------------------------------------------------------------------------
Primary Ports
  Inputs                    19           0            0     19    100.00%
  Outputs                    9           0            0      9    100.00%
  I/O                        0           0            0      0        N/A
Sequential Outputs
  Memory                     0           0            0      0        N/A
  Flop                       9           0            0      9    100.00%
  Latch                      0           0            0      0        N/A
  Arch ICGC                  0           0            0      0        N/A
  Inferred ICGC              0           0            0      0        N/A
  Total ICGC                 0           0            0      0        N/A
Drivers
  Driver nets               44           0            0     44    100.00%
  RTL Driver nets            9           0            0      9    100.00%
DFT
  Input Ports                0           0            0      0        N/A
  Flop Outputs               0           0            0      0        N/A
  Memory Outputs             0           0            0      0        N/A
---------------------------------------------------------------------------
```

Lab 06

Leitura do VCD e análise de potência (GENUS)

Comando(s) para o arquivo “${DESIGNS}.tcl”:

set_db power_engine joules ;# <joules or legacy>

report_sdb_annotation ;# -show_details unasserted -inputs

report_power -unit uW

Para o projeto “somador”:

- Popular a tabela abaixo:

freq MHz (clk)	testbench	power µW			lp_computed_probability (nets)		*total area* μm^2
		leakage	dynamic	total	a_i[7]	sum_o[7]	
10	--						
	X						
	2X						
100	--						
	X						
	2X